Chap 1 考点

1. pure substance, solution, emulsion & suspension

- pure substance { elements O_2, Au / compounds H_2O

- solution (homogeneous mixture): x filtration √ distillation 蒸馏 & crystallation 结晶 乙醇+水
  eg. salt water, sugar in tea, sea water 可溶盐+水, vinegar (acetic acid in water) 冰醋酸+水, beer, alloy 合金

- emulsion: a special type of mixture of 2 immiscible liquids
  eg. milk (l→l), cream (g→l), salad dressings, fog (l→g), smoke (s→g), marshmallow (g→s), butter (l→s), paint (s→l), opal (s→s)

- suspension: solid particles are dispersed in a liquid (or gas)
  eg. muddy water 有泥沙

2. ex/intensive → x vary with n. 广度 强度
   ↳ vary with n.

3. Quantum Numbers n, l & m_l

能级 n	principle quantum number 主量子数	1, 2, 3 …	the energy level of electrons (能层)
l	azimuthal quantum number 角量子数	一定记得0开始 $\boxed{0}$, 1, 2 … $\boxed{(n-1)}$ → s p d f g h …	the distribution of electron cloud in space
m_l	magnetic quantum number 磁量子数	0, ±1, ±2 … $\boxed{\pm l}$	orientation of the orbital

n	l	m_l
1	0	0
2	0	0
	1	0, ±1
3	0 3s	0
	1 3p	0, ±1
	2 3d	0, ±1, ±2

2l+1 possibilities

(1, 0, 0) 1s
(5, 4, −1) 5g
从0开始 (4, 3, 3) 4f
(2, 0, 1) X

n → 2 ← l, p_x ← m_l ($2p_x$)

m_s electron spin number 自旋磁量子数 $\pm\frac{1}{2}$

He($1s^2$)
- one electron has the quantum number $n=1, l=0, m_l=0, m_s=+1/2$
- the second electron: $n=1, l=0, m_l=0, m_s=-1/2$

4. $^{0}_{-1}e$ $^{1}_{1}p$ $^{1}_{0}n$ (neutron) →质量num
   →描述这种粒子本身的电荷属性, 和原子核里的质子两回事

5. 价层: valence shell
   完整电子排布式: full electron configuration $ns<(n-2)f<(n-1)d<np$
   简化: condensed / Noble gas notation 电子一定是从外向内失去
   电子排布图: orbital occupancy diagram / orbital filling diagram
   原子结构示意图: atomic structure diagram / Bohr Model of the atom
   同位素符号: $^{A}_{Z}X$ isotope symbol

决定电荷数
6pro 7n 5e $^{13}_{6}C^{+}$
↳决定元素

eg. Mn^{2+}
for valence shell { electron configuration $3d^5$ / orbital occupancy diagram ↑|↑|↑|↑|↑

fec: Au^{3+} $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^{10} 4f^{14} 5s^2 5p^6 5d^8$ 从外向内失电子

6. common techniques to separate mixtures
- filtration 过滤 insoluble solid vs solvent
- distillation 蒸馏 liquids with different boiling points
- crystallization 结晶 soluble solid vs solvent
- solvent extraction 溶剂萃取 immiscible liquids → 不互溶的
- Sublimation 升华

扫描全能王 3亿人都在用的扫描App

Chap 2 考点

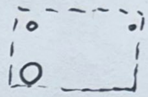
noble gases
group 族
period 周期
transition medal
lanthanides
actinides

metalloid 类金属
alkali metal 碱金属 G1 (除H)
alkaline earth metal 碱土金属 G2
halogen 卤素 G17
periodicity
W: Tungsten
Pt: Platinum

U: Uranium
redox reaction 氧还
conproportionation reaction 归中
dis 歧化
peroxide O_2^{2-}
superoxide O_2^- KO_2
allotropes 同素异形体
C_2^{2-} carbide

1. Zeff: the actual magnitude of positive charge that is experienced by an electron in the atom $= Z$ (the nuclear charge) $- \sigma$ (shielding constant)
↑ (the number of protons)
↑(少)

2. Radii $\propto \frac{n^2}{Z_{eff}}$ · → n Zeff↑ R↓
   · ↓ Zeff↑少, n^2 increases more ∴ R↑



n: 原子层数

3. electronic ~~on~~ configuration $ns<(n-2)f<(n-1)d<np$ 能

3d与4s { 填入时: 先填4s后填3d
书写时: 3d写在4s前

过渡金属失去e^-时: 先失4s后3d 电子一定是从外向内失

4. empirical formula 实验式 (最简整数比) molecular formula 分子式
combustion analysis 燃烧分析法

5. Vm ⇒ 22.4L/mol at s.t.p (standard temperature & pressure. 0°C, 1bar)
1bar $=10^5$ Pa

6 看清是 { chemical / full } equation vs ionic equation

7. H: most abundant in the universe
$MH_n + H_2O \rightarrow H_2$ & base
protium, deuterium, tritium

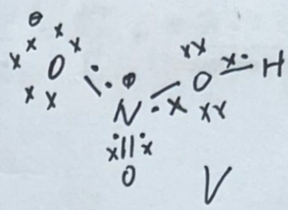
8. C: organic compounds: alkane, alkene, alkyne, alcohol, carboxylic acid 羧酸

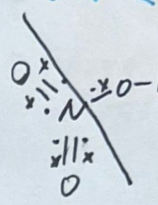
Chap 3 考点

1. inphase overlap　　out of phase overlap

2. Lewis structure

无d轨道

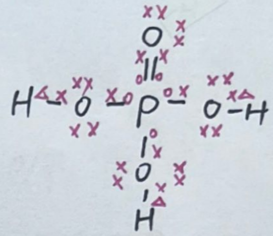




Dot-and-cross structure

- bp
- lp
- bond order : 1 single bond　　2 double bond

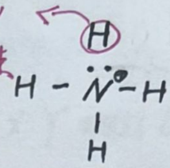
有d轨道



Octet rule

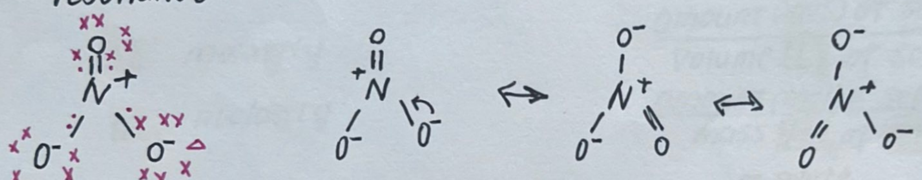
- exception: H 2
- exception: B 6 & Be 4
- exception: hypervalent (有d轨道)

本来带正电，得了e^-变中性



- dative bond (coordinate bond) 配位键 箭头是从提供孤电子对的原子指向接受孤电子对的原子.
- formal charge 形式电荷 = 该原子的价电子数 − (孤电子数 + 成键电子数/2)

3. resonance

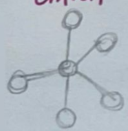


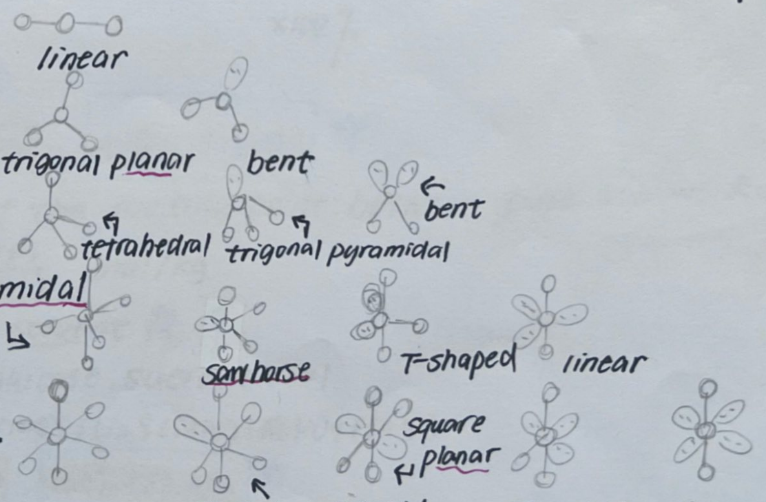
resonance hybrid

4. VSEPR　价VSEPR : electron pair geometry (including lp)

实际构型 : molecular geometry　(ex‥　　)

	VSEPR	lone pairs: 0	1	2	3	4
2 sp	linear	linear				
3 sp^2	trigonal planar	trigonal planar	bent			
4 sp^3	tetrahedral	tetrahedral	trigonal pyramidal	bent		
5	trigonal bipyramidal		seesaw	T-shaped	linear	
6	Octahedral		square pyramidal	square planar	T-shaped	linear





5. $Mg^{2+}(g) + 2F^-(g) \rightarrow MgF_2(s) + LE$ → lattice energy $\propto \frac{q_1 q_2}{r_1 + r_2}$　r影响比charge小很多　先比q再比r

Chap 4 考点 丁零加油

1. $1atm = 760mmHg = 1.01325 \times 10^5 Pa = 101.325 kPa = 760 torr$ $1 bar = 10^5 Pa$

狠: PV 查: V8T

$PV = nRT \rightarrow K$

Pa m^3 mol $\rightarrow 8.314 J \cdot mol^{-1} \cdot K^{-1}$

$\rho = \frac{m}{V} = \frac{n \cdot M \cdot P}{nRT}$ ($m \rightarrow g$, $M \rightarrow g/mol$, $V \rightarrow m^3$) $\rho \propto M$

$V_m = \frac{V}{n} = \frac{RT}{P} = \frac{8.314 \times 298K}{1.01325 \times 10^5 Pa} = 0.02445 m^3/mol$ ($\rightarrow 25°C$)

$24.45 L/mol$

CO_2 in air is 420 ppm

parts per million 百万分之一

$101325 \cdot \frac{420}{10^6} = 42.5565 Pa$

$P_A = \chi_A \times P_{total}$

↳ the partial pressure

2. solute ~质 solvent ~剂

hydrogen bond X–H···Y linear! 180° X, Y must be N/O/F

exothermic 放 endothermic 吸

T↑ gas solubility↓

看清题目算的溶质还是溶液质量 (HM).

3. Quantitative ways of expressing c

M molarity	$\frac{\text{amount (mol) of solute}}{\text{volume (L) of solution}}$
m molality	$\frac{\text{amount (mol) of solute}}{\text{mass (kg) of solvent}}$ ☆
mass percent	$\frac{m_{solute}}{m_{solution}} \times 100\%$
volume percent	$\frac{V_{solute}}{V_{solution}} \times 100\%$
mole fraction χ	$\frac{\text{amount (mol) of solute}}{\text{amount (mol) of solute + amount (mol) of solvent}}$
mole percent mol%	$\times 100\%$

4. Colligative properties

— Freezing-point Depression $\boxed{\Delta T_f = iK_f m}$ ☆

· ΔT_f: the difference of the melting point between pure solvent & solution

· m: molality ‼ $\frac{n(solute)}{m(solvent)}$ mol/kg

(cryoscopic constant) K_f: only dependent on solvent ☆

(van't Hoff factor) · i: non-electrolyte (glucose, sucrose) i=1

electrolyte: NaCl (i=2); Li_2S (i=3); K_3PO_4 (i=4)

— Boiling-point Elevation $\boxed{\Delta T_b = iK_b m}$ ☆

↳ only dependent on the solvent (ebullioscopic constant)

5. types of colloids

Colloid type	dispersed substance ··	medium	examples
aerosol	l	g	fog
aerosol	s	g	smoke
foam	g	l	whipped cream
solid foam	g	s	marshmellow
emulsion	l	l	milk
solid emulsion	l	s	butter
sol (溶胶)	s	l	paint ; cell fluid
solid sol	s	s	opal

Chap 5 考点

1. rate is defined as a $\boxed{\text{positive}}$ quantity

Δ: average
d: instantaneous

2. $aA+bB \to cC+dD$

$r = -\frac{1}{a}\frac{d[A]}{dt} = \frac{1}{c}\frac{d[C]}{dt}$

- divide by stoichiometric number
- positive for product ; negative for reactant
- in any case, the rate must be a positive valut

3. beaker 烧杯
test tube 试管
conical flask 锥形瓶
volumetric flask 容量瓶
round bottom flask 圆底烧瓶
burette 滴定管
pipette 移液管

4. rate law:

整个化学反应方程式的 rate

$aA+bB \to cC+dD$ $\quad$ $\text{Rate} = k[A]^m[B]^n$

$mol \cdot L^{-1} \cdot s^{-1}$ $\quad$ $(mol \cdot L^{-1})^n$

the rate constant

mth order with respect to A
overall order : (m+n)th order

Decimal places (Dp, dp) eg. 2dp 2345.00 6.95

Significant figure (sig fig)

5. $t_{1/2}$ Half-life: C反减少到初始值一半所需时间 $[A]_t \to \frac{[A]_0}{2}$

— zero-order reaction

$\text{rate} = -\frac{d[A]}{dt} = k_0[A]^0$ $\quad$ $d[A] = -k_0 dt$ $\quad$ $\boxed{[A]} = -k_0 t + [A_0]$ $\quad$ unit: $mol \cdot L^{-1} \cdot s^{-1}$

— first-order reaction

$\text{rate} = -\frac{d[A]}{dt} = k_1[A]^1$ $\quad$ $\frac{d[A]}{[A]} = -k_1 dt$ $\quad$ $\boxed{\ln[A]} = -k_1 t + \ln[A]_0$ $\quad$ s^{-1}

— second-order

$\text{rate} = -\frac{d[A]}{dt} = k_2[A]^2$ $\quad$ $\frac{d[A]}{[A]^2} = -k_2 dt$ $\quad$ $\boxed{[A]^{-1}} = k_2 t + [A_0]^{-1}$ $\quad$ $L \cdot mol^{-1} \cdot s^{-1}$

1M/s = 60M/min = 3600M/h
看清换算的是啥.

6. Arrhenius equation

$k = Ae^{-Ea/RT}$ $\quad$ $8.314 J \cdot mol^{-1} \cdot K^{-1}$ $\quad$ Ea: J/mol

$\ln k = \ln A - \frac{Ea}{RT}$ $\quad$ $\ln\frac{k_2}{k_1} = \frac{Ea}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$ $\quad$ enzymes 酶

用对数消去指数，用消 lnA 去联立求解